

Student Management System | To develop a console-based St... | Online C++ Compiler - Progra... |

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C++ Online Compiler

main.cpp

```
1 #include <iostream>
2 #include <fstream>
3 #include <string>
4 #include <vector>
5
6 using namespace std;
7
8 struct Student {
9     int id;
10    string name;
11    float gpa;
12 };
13
14 void addStudent() {
15     Student s;
16     ofstream outfile("students.txt", ios::app);
17
18     cout << "Enter ID: "; cin >> s.id;
19     cin.ignore();
20     cout << "Enter Name: "; getline(cin, s.name);
21     cout << "Enter GPA: "; cin >> s.gpa;
22
23     outfile << s.id << " " << s.name << " " << s.gpa << endl;
24     outfile.close();
25     cout << "Record saved successfully!\n";
26 }
27
28 void displayStudents() {
29     Student s;
30     ifstream inFile("students.txt");
31     cout << "\n--- Student Records ---\n";
32     while (inFile >> s.id >> s.name >> s.gpa) {
33         cout << "ID: " << s.id << " | Name: " << s.name << " | GPA: " << s.gpa << endl;
34     }
35     inFile.close();
36 }
37
```

Output

```
1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 1
Enter ID: 30
Enter Name: tanuja
Enter GPA: 7.6
Record saved successfully!

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 3
Enter ID to delete: 30
Record not found.

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 4

--- Code Execution Successful ---
```

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main.cpp

```
38 void deleteStudent() {
39     int targetId;
40     cout << "Enter ID to delete: "; cin >> targetId;
41
42     ifstream inFile("students.txt");
43     ofstream tempFile("temp.txt");
44     Student s;
45     bool found = false;
46
47     while (inFile >> s.id >> s.name >> s.gpa) {
48         if (s.id != targetId) {
49             tempFile << s.id << " " << s.name << " " << s.gpa << endl;
50         } else {
51             found = true;
52         }
53     }
54     inFile.close();
55     tempFile.close();
56
57     remove("students.txt");
58     rename("temp.txt", "students.txt");
59
60     if (found) cout << "Record deleted.\n";
61     else cout << "Record not found.\n";
62 }
63
64 int main() {
65     int choice;
66     while (true) {
67         cout << "\n1. Add Student\n2. Display All\n3. Delete Student\n4. Exit\nChoice: ";
68         cin >> choice;
69
70         switch (choice) {
71             case 1: addStudent(); break;
72             case 2: displayStudents(); break;
73             case 3: deleteStudent(); break;
74             case 4: return 0;
75         }
76     }
77 }
```

Output

```
1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 1
Enter ID: 30
Enter Name: tanuja
Enter GPA: 7.6
Record saved successfully!

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 2

--- Student Records ---

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 3
Enter ID to delete: 30
Record not found.

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 4

--- Code Execution Successful ---
```

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Student Management System

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Programiz PRO

main.cpp

Run

Clear

```
42 ifstream inFile("students.txt");
43 ofstream outFile("temp.txt");
44 Student s;
45 bool found = false;
46
47 while (inFile >> s.id >> s.name >> s.gpa) {
48     if (s.id != targetId) {
49         outFile << s.id << " " << s.name << " " << s.gpa << endl;
50     } else {
51         found = true;
52     }
53 }
54 inFile.close();
55 outFile.close();
56
57 remove("students.txt");
58 rename("temp.txt", "students.txt");
59
60 if (found) cout << "Record deleted.\n";
61 else cout << "Record not found.\n";
62
63
64 int main() {
65     int choice;
66     while (true) {
67         cout << "\n1. Add Student\n2. Display All\n3. Delete Student\n4. Exit\nChoice: ";
68         cin >> choice;
69
70         switch (choice) {
71             case 1: addStudent(); break;
72             case 2: displayStudents(); break;
73             case 3: deleteStudent(); break;
74             case 4: return 0;
75             default: cout << "Invalid choice!\n";
76         }
77     }
78 }
```

Output

```
1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 1
Enter ID: 30
Enter Name: tanuja
Enter GPA: 7.6
Record saved successfully!

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 2

--- Student Records ---

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 3
Enter ID to delete: 30
Record not found.

1. Add Student
2. Display All
3. Delete Student
4. Exit
Choice: 4

=== Code Execution Successful ===
```

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